

Take Home Exam 1 - Healthcare Data

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Instructions Download the datasets and load them into R: messy_health_records.csv, patients.csv, and medications.csv

Answer all questions. Provide explanations alongside your R code. Ensure clear and well-commented R code. Your code should be reproducible. Include tables, graphs, and visualizations where applicable. Cite sources if using external packages beyond tidyverse. Submit a PDF onto Gradescope

A PDF document with responses, R code, and interpretations and R appendix. A .R script file containing all R code may be requested during grading. Do not submit before you get a request from teaching staff.

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

Import Dataset

```
# import the datasets

# messy_health_records.csv -> messy_health_records
messy_health_records <- read_csv("messy_health_records.csv")

## Rows: 8670 Columns: 9
## -- Column specification -----
## Delimiter: ","
## chr  (4): condition, blood_pressure, medication_given, hospital_location
## dbl  (4): record_id, patient_id, cholesterol_level, patient_age
## date (1): visit_date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.

# patients.csv -> patients
patients <- read_csv("patients.csv")

## Rows: 3513 Columns: 5
## -- Column specification -----
```

```
## Delimiter: ","
## chr (3): patient_name, gender, insurance_status
## dbl (1): patient_id
## date (1): admission_date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
# medications.csv -> medications
medications <- read_csv("medications.csv")
```

```
## Rows: 5 Columns: 3
## -- Column specification -----
## Delimiter: ","
## chr (2): condition, recommended_medication
## dbl (1): dosage_mg
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# briefly know the structures of the three csv files with head()
head(messy_health_records)
```

```
## # A tibble: 6 x 9
##   record_id patient_id visit_date condition blood_pressure cholesterol_level
##   <dbl>      <dbl> <date>      <chr>      <chr>                <dbl>
## 1      367      1194 2020-01-16 Hypertension Normal                184
## 2      7908     4911 2020-11-25 Obesity     Elevated              128.
## 3      2115     1530 2020-03-29 Hypertension Normal                155.
## 4      1545     3840 2020-03-05 Obesity     Stage 1 Hypert~      237.
## 5      7028     1178 2020-10-19 Diabetes     Elevated              178
## 6      6493     1893 2020-09-27 Hypertension Normal                247.
## # i 3 more variables: medication_given <chr>, hospital_location <chr>,
## #   patient_age <dbl>
```

```
head(patients)
```

```
## # A tibble: 6 x 5
##   patient_id patient_name gender admission_date insurance_status
##   <dbl> <chr>      <chr> <date>      <chr>
## 1      1194 Patient_0    Female 2015-01-01    Medicaid
## 2      4911 Patient_1    Female 2015-01-02    Private
## 3      1530 Patient_2    Male   2015-01-03    Medicare
## 4      3840 Patient_3    Other  2015-01-04    Medicaid
## 5      1178 Patient_4    Female 2015-01-05    Private
## 6      1893 Patient_5    Male   2015-01-06    Private
```

```
head(medications)
```

```
## # A tibble: 5 x 3
##   condition      recommended_medication dosage_mg
##   <chr>          <chr>                <dbl>
## 1 Hypertension Lisinopril              10
## 2 Diabetes     Metformin             500
## 3 Asthma       Albuterol              90
## 4 Heart Disease Atorvastatin           20
## 5 Obesity      Orlistat              120
```

Question 1: Data Cleaning and Preprocessing (20 points)

Using the Messy Health Records Dataset, perform the following tasks:

1. Check for missing values and provide a summary.
2. Handle missing values (e.g., impute, remove, or fill) and justify your choices.
3. Convert incorrect data types, ensuring visit_date is in Date format.
4. Remove duplicate records, if any. Show how many were found and removed.
5. Detect and handle outliers in cholesterol_level and patient_age using the IQR method.

Deliverables: - R code for cleaning - Summary tables before and after cleaning - Justifications for all cleaning decisions

```
# Here is Part 1 of Question 1.
# check for missing values by counting the numbers of missing values in all columns
missing_values_summary_11 <- colSums(is.na(messy_health_records))
missing_values_summary_11
```

```
##      record_id      patient_id      visit_date      condition
##           0           0           0           0
## blood_pressure cholesterol_level medication_given hospital_location
##          437          439           0           0
##      patient_age
##           433
```

```
# Here is Part 2 of Question 1.
# use drop_na() to remove those rows with missing values
# there are thousands of records, it is okay to drop those rows with NA
messy_health_records_12 <- messy_health_records %>% drop_na()
messy_health_records_12
```

```
## # A tibble: 7,431 x 9
##   record_id patient_id visit_date condition blood_pressure cholesterol_level
##   <dbl>      <dbl> <date>      <chr>      <chr>              <dbl>
## 1     7908     4911 2020-11-25 Obesity    Elevated          128.
## 2     2115     1530 2020-03-29 Hypertension Normal          155.
## 3     1545     3840 2020-03-05 Obesity    Stage 1 Hyper~    237.
## 4     7028     1178 2020-10-19 Diabetes    Elevated          178
## 5     6493     1893 2020-09-27 Hypertension Normal          247.
## 6     5216     4292 2020-08-05 Diabetes    Stage 1 Hyper~    151.
## 7     6388     4426 2020-09-23 Heart Disea~ Stage 2 Hyper~    193.
## 8       412     3595 2020-01-18 Obesity    Elevated          220.
## 9     4202     1062 2020-06-24 Asthma      Elevated          220.
## 10    7500     1336 2020-11-08 Diabetes    Normal           126.
## # i 7,421 more rows
## # i 3 more variables: medication_given <chr>, hospital_location <chr>,
## #   patient_age <dbl>
```

```
# Here is Part 3 of Question 1.
# use ymd() in lubridate package to convert visit_date into Date format
messy_health_records_13 <- messy_health_records_12 %>%
  mutate(visit_date = ymd(visit_date))
messy_health_records_13
```

```
## # A tibble: 7,431 x 9
```

```
##      record_id patient_id visit_date condition    blood_pressure cholesterol_level
##      <dbl>      <dbl> <date>      <chr>      <chr>              <dbl>
## 1      7908      4911 2020-11-25 Obesity      Elevated          128.
## 2      2115      1530 2020-03-29 Hypertension Normal           155.
## 3      1545      3840 2020-03-05 Obesity      Stage 1 Hyper~    237.
## 4      7028      1178 2020-10-19 Diabetes      Elevated          178
## 5      6493      1893 2020-09-27 Hypertension Normal           247.
## 6      5216      4292 2020-08-05 Diabetes      Stage 1 Hyper~    151.
## 7      6388      4426 2020-09-23 Heart Disea~ Stage 2 Hyper~    193.
## 8       412      3595 2020-01-18 Obesity      Elevated          220.
## 9      4202      1062 2020-06-24 Asthma      Elevated          220.
## 10     7500      1336 2020-11-08 Diabetes      Normal            126.
## # i 7,421 more rows
## # i 3 more variables: medication_given <chr>, hospital_location <chr>,
## #   patient_age <dbl>
```

Here is Part 4 of Question 1.

remove those repetitive rows by using distinct()

```
messy_health_records_14 <- messy_health_records_13 %>% distinct()
messy_health_records_14
```

```
## # A tibble: 7,291 x 9
##      record_id patient_id visit_date condition    blood_pressure cholesterol_level
##      <dbl>      <dbl> <date>      <chr>      <chr>              <dbl>
## 1      7908      4911 2020-11-25 Obesity      Elevated          128.
## 2      2115      1530 2020-03-29 Hypertension Normal           155.
## 3      1545      3840 2020-03-05 Obesity      Stage 1 Hyper~    237.
## 4      7028      1178 2020-10-19 Diabetes      Elevated          178
## 5      6493      1893 2020-09-27 Hypertension Normal           247.
## 6      5216      4292 2020-08-05 Diabetes      Stage 1 Hyper~    151.
## 7      6388      4426 2020-09-23 Heart Disea~ Stage 2 Hyper~    193.
## 8       412      3595 2020-01-18 Obesity      Elevated          220.
## 9      4202      1062 2020-06-24 Asthma      Elevated          220.
## 10     7500      1336 2020-11-08 Diabetes      Normal            126.
## # i 7,281 more rows
## # i 3 more variables: medication_given <chr>, hospital_location <chr>,
## #   patient_age <dbl>
```

count the number of total rows after finishing Part 3 with nrow()

```
row_count <- nrow(messy_health_records_13)
```

count the number of unique rows after finishing Part 3 with n_distinct()

```
unique_row_count <- n_distinct(messy_health_records_13)
```

the number of duplicate rows is just the difference between row_count and unique_row_count

```
duplicates <- row_count - unique_row_count
duplicates
```

```
## [1] 140
```

Here is Part 5 of Question 1.

calculate the upper and lower quantiles of cholesterol_level with quantile()

the lower quartile is 0.25

```
cho25 <- quantile(messy_health_records_14$cholesterol_level, 0.25, na.rm = TRUE)
```

the upper quartile is 0.75

```
cho75 <- quantile(messy_health_records_14$cholesterol_level, 0.75, na.rm = TRUE)
```

Interquartile range = upper quartile - lower quartile

```
IQR_cho <- cho75 - cho25
```

```

# consider the outliers as 1.5 interquartile range smaller than cho25 or greater than cho75
lower_bound_cho <- cho25 - 1.5 * IQR_cho
upper_bound_cho <- cho75 + 1.5 * IQR_cho

# calculate the upper and lower quartiles of patient_age with quantile()
# the lower quartile is 0.25
age25 <- quantile(messy_health_records_14$patient_age, 0.25, na.rm = TRUE)
# the upper quartile is 0.75
age75 <- quantile(messy_health_records_14$patient_age, 0.75, na.rm = TRUE)
# Interquartile range = upper quartile - lower quartile
IQR_age <- age75 - age25
# consider the outliers as 1.5 interquartile range smaller than cho25 or greater than cho75
lower_bound_age <- age25 - 1.5 * IQR_age
upper_bound_age <- age75 + 1.5 * IQR_age

# use filter() to remain those ordinary records
messy_health_records_15 <- messy_health_records_14 %>%
  # normal cholesterol_level should be between lower_bound_cho and upper_bound_cho
  filter(cholesterol_level >= lower_bound_cho & cholesterol_level <= upper_bound_cho) %>%
  # normal patient_age should be between lower_bound_age and upper_bound_age
  filter(patient_age >= lower_bound_age & patient_age <= upper_bound_age)
messy_health_records_15

## # A tibble: 7,291 x 9
##   record_id patient_id visit_date condition    blood_pressure cholesterol_level
##   <dbl>      <dbl> <date>      <chr>      <chr>              <dbl>
## 1      7908      4911 2020-11-25 Obesity    Elevated          128.
## 2      2115      1530 2020-03-29 Hypertension Normal          155.
## 3      1545      3840 2020-03-05 Obesity    Stage 1 Hyper~    237.
## 4      7028      1178 2020-10-19 Diabetes    Elevated          178
## 5      6493      1893 2020-09-27 Hypertension Normal          247.
## 6      5216      4292 2020-08-05 Diabetes    Stage 1 Hyper~    151.
## 7      6388      4426 2020-09-23 Heart Disea~ Stage 2 Hyper~    193.
## 8       412      3595 2020-01-18 Obesity    Elevated          220.
## 9      4202      1062 2020-06-24 Asthma     Elevated          220.
## 10     7500      1336 2020-11-08 Diabetes    Normal           126.
## # i 7,281 more rows
## # i 3 more variables: medication_given <chr>, hospital_location <chr>,
## #   patient_age <dbl>

```

Explanation for part 2: There are thousands of records in this tibble, dropping those rows with NA will not greatly affect the result and conclusion of the data analysis.

Question 2: Data Transformation & Feature Engineering (20 points)

After cleaning the dataset, perform the following transformations:

1. Create a new column `visit_month` by extracting the month from `visit_date`.
2. Create a categorical column `patient_age_group` that groups patients into: “Young” (18-35) “Middle-aged” (36-60) “Senior” (61+)
3. Create a new column `cholesterol_risk`, categorizing levels as: “Low” (<160 mg/dL), “Moderate” (160-200 mg/dL), “High” (>200 mg/dL)
4. Summarize the number of visits per hospital location and condition using a pivot function.

Deliverables: - R code for transformations - A table showing new columns and pivoted data - Explanation of transformations and insights

```
# Here is Part 1 of Question 2.
# continued from Part 5 of Question 1
messy_health_records_21 <- messy_health_records_15 %>%
  # use month() in lubridate package and mutate() to add the new column
  mutate(visit_month = month(visit_date))
messy_health_records_21

## # A tibble: 7,291 x 10
##   record_id patient_id visit_date condition    blood_pressure cholesterol_level
##   <dbl>      <dbl> <date>      <chr>      <chr>              <dbl>
## 1      7908      4911 2020-11-25 Obesity    Elevated          128.
## 2      2115      1530 2020-03-29 Hypertension Normal          155.
## 3      1545      3840 2020-03-05 Obesity    Stage 1 Hyper~    237.
## 4      7028      1178 2020-10-19 Diabetes    Elevated          178
## 5      6493      1893 2020-09-27 Hypertension Normal          247.
## 6      5216      4292 2020-08-05 Diabetes    Stage 1 Hyper~    151.
## 7      6388      4426 2020-09-23 Heart Disea~ Stage 2 Hyper~    193.
## 8       412      3595 2020-01-18 Obesity    Elevated          220.
## 9      4202      1062 2020-06-24 Asthma     Elevated          220.
## 10     7500      1336 2020-11-08 Diabetes    Normal           126.
## # i 7,281 more rows
## # i 4 more variables: medication_given <chr>, hospital_location <chr>,
## #   patient_age <dbl>, visit_month <dbl>
```

```
# Here is Part 2 of Question 2.
# add the new column called "patient_age_group" with mutate()
messy_health_records_22 <- messy_health_records_21 %>%
  # there are 3 cases. Use case_when() to separate the 3 cases.
  mutate(patient_age_group = case_when(
    patient_age >= 18 & patient_age <= 35 ~ "Young",
    patient_age >= 36 & patient_age <= 60 ~ "Middle-aged",
    patient_age >= 61 ~ "Senior"))
messy_health_records_22
```

```
## # A tibble: 7,291 x 11
##   record_id patient_id visit_date condition    blood_pressure cholesterol_level
##   <dbl>      <dbl> <date>      <chr>      <chr>              <dbl>
## 1      7908      4911 2020-11-25 Obesity    Elevated          128.
## 2      2115      1530 2020-03-29 Hypertension Normal          155.
## 3      1545      3840 2020-03-05 Obesity    Stage 1 Hyper~    237.
## 4      7028      1178 2020-10-19 Diabetes    Elevated          178
## 5      6493      1893 2020-09-27 Hypertension Normal          247.
## 6      5216      4292 2020-08-05 Diabetes    Stage 1 Hyper~    151.
## 7      6388      4426 2020-09-23 Heart Disea~ Stage 2 Hyper~    193.
## 8       412      3595 2020-01-18 Obesity    Elevated          220.
## 9      4202      1062 2020-06-24 Asthma     Elevated          220.
## 10     7500      1336 2020-11-08 Diabetes    Normal           126.
## # i 7,281 more rows
## # i 5 more variables: medication_given <chr>, hospital_location <chr>,
## #   patient_age <dbl>, visit_month <dbl>, patient_age_group <chr>
```

```
# Here is Part 3 of Question 2.
# similarly, add the new column called "cholesterol_risk" with mutate()
```

```

messy_health_records_23 <- messy_health_records_22 %>%
  # similarly, there are 3 cases. Use case_when() to separate the 3 cases.
  mutate(cholesterol_risk = case_when(
    cholesterol_level < 160 ~ "Low",
    cholesterol_level >= 160 & cholesterol_level <= 200 ~ "Moderate",
    cholesterol_level > 200 ~ "High"))
messy_health_records_23

## # A tibble: 7,291 x 12
##   record_id patient_id visit_date condition    blood_pressure cholesterol_level
##   <dbl>      <dbl> <date>      <chr>      <chr>              <dbl>
## 1      7908      4911 2020-11-25 Obesity    Elevated          128.
## 2      2115      1530 2020-03-29 Hypertension Normal          155.
## 3      1545      3840 2020-03-05 Obesity    Stage 1 Hyper~    237.
## 4      7028      1178 2020-10-19 Diabetes    Elevated          178.
## 5      6493      1893 2020-09-27 Hypertension Normal          247.
## 6      5216      4292 2020-08-05 Diabetes    Stage 1 Hyper~    151.
## 7      6388      4426 2020-09-23 Heart Disea~ Stage 2 Hyper~    193.
## 8       412      3595 2020-01-18 Obesity    Elevated          220.
## 9      4202      1062 2020-06-24 Asthma      Elevated          220.
## 10     7500      1336 2020-11-08 Diabetes    Normal           126.
## # i 7,281 more rows
## # i 6 more variables: medication_given <chr>, hospital_location <chr>,
## #   patient_age <dbl>, visit_month <dbl>, patient_age_group <chr>,
## #   cholesterol_risk <chr>

```

```

# Here is Part 4 of Question 2.
messy_health_records_24 <- messy_health_records_23 %>%
  # per hospital location and condition means we need to use group_by() to separate
  group_by(hospital_location, condition) %>%
  # count the number of patients per hospital with the same condition
  summarize(visit_count = n()) %>%
  # change the NA to 0 with replace_na()
  # mutate should be used to help keep the list structure
  mutate(visit_count = replace_na(visit_count, 0)) %>%
  # convert the data frame to wider form with pivot_wider()
  # names_from and values_from are two parameters
  pivot_wider(names_from = condition, values_from = visit_count)

```

```

## `summarise()` has grouped output by 'hospital_location'. You can override using
## the `.groups` argument.

```

```

messy_health_records_24

## # A tibble: 5 x 6
## # Groups:   hospital_location [5]
##   hospital_location Asthma Diabetes `Heart Disease` Hypertension Obesity
##   <chr>             <int>   <int>         <int>         <int>   <int>
## 1 Chicago          297     320           290           327     308
## 2 Houston          295     283           304           256     305
## 3 LA               284     292           271           303     321
## 4 Miami            297     282           274           288     266
## 5 NYC              263     318           279           287     281

```

Question 3: Data Joining and Merging (20 points)

Using the Patient Data and Medication Data, perform the following:

1. Merge the patient dataset using patient_id and the health records dataset.
2. Merge the medication dataset using condition and the health records dataset.
3. Filter out records where dosage_mg exceeds 200 mg (possible data inconsistencies).
4. Identify the top 5 most frequently treated conditions, showing their total patient count.

Deliverables: - R code for merging datasets - A table showing the merged dataset structure - The top 5 most common conditions

```
# Here is Part 1 of Question 3.
# continued from Part 3 of Question 2
health_record_31 <- messy_health_records_23 %>%
  # use left_join() to join the two datasets according to the "patient_id" column
  left_join(patients, by = "patient_id")
health_record_31
```

```
## # A tibble: 7,291 x 16
##   record_id patient_id visit_date condition blood_pressure cholesterol_level
##   <dbl>      <dbl> <date>      <chr>      <chr>              <dbl>
## 1      7908      4911 2020-11-25 Obesity     Elevated           128.
## 2      2115      1530 2020-03-29 Hypertension Normal            155.
## 3      1545      3840 2020-03-05 Obesity     Stage 1 Hyper~    237.
## 4      7028      1178 2020-10-19 Diabetes    Elevated           178
## 5      6493      1893 2020-09-27 Hypertension Normal            247.
## 6      5216      4292 2020-08-05 Diabetes    Stage 1 Hyper~    151.
## 7      6388      4426 2020-09-23 Heart Disea~ Stage 2 Hyper~    193.
## 8       412      3595 2020-01-18 Obesity     Elevated           220.
## 9      4202      1062 2020-06-24 Asthma      Elevated           220.
## 10     7500      1336 2020-11-08 Diabetes    Normal             126.
## # i 7,281 more rows
## # i 10 more variables: medication_given <chr>, hospital_location <chr>,
## #   patient_age <dbl>, visit_month <dbl>, patient_age_group <chr>,
## #   cholesterol_risk <chr>, patient_name <chr>, gender <chr>,
## #   admission_date <date>, insurance_status <chr>
```

```
# Here is Part 2 of Question 3.
health_record_32 <- health_record_31 %>%
  # use left_join() to join the two datasets according to the "condition" column
  left_join(medications, by = "condition")
health_record_32
```

```
## # A tibble: 7,291 x 18
##   record_id patient_id visit_date condition blood_pressure cholesterol_level
##   <dbl>      <dbl> <date>      <chr>      <chr>              <dbl>
## 1      7908      4911 2020-11-25 Obesity     Elevated           128.
## 2      2115      1530 2020-03-29 Hypertension Normal            155.
## 3      1545      3840 2020-03-05 Obesity     Stage 1 Hyper~    237.
## 4      7028      1178 2020-10-19 Diabetes    Elevated           178
## 5      6493      1893 2020-09-27 Hypertension Normal            247.
## 6      5216      4292 2020-08-05 Diabetes    Stage 1 Hyper~    151.
## 7      6388      4426 2020-09-23 Heart Disea~ Stage 2 Hyper~    193.
## 8       412      3595 2020-01-18 Obesity     Elevated           220.
## 9      4202      1062 2020-06-24 Asthma      Elevated           220.
## 10     7500      1336 2020-11-08 Diabetes    Normal             126.
```



```
## # i 7,281 more rows
## # i 12 more variables: medication_given <chr>, hospital_location <chr>,
## #   patient_age <dbl>, visit_month <dbl>, patient_age_group <chr>,
## #   cholesterol_risk <chr>, patient_name <chr>, gender <chr>,
## #   admission_date <date>, insurance_status <chr>,
## #   recommended_medication <chr>, dosage_mg <dbl>

# Here is Part 3 of Question 3.
health_record_33 <- health_record_32 %>%
  # use filter() to remain those rows we need here
  filter(dosage_mg <= 200)
health_record_33

## # A tibble: 5,796 x 18
##   record_id patient_id visit_date condition    blood_pressure cholesterol_level
##   <dbl>      <dbl> <date>      <chr>      <chr>              <dbl>
## 1      7908      4911 2020-11-25 Obesity    Elevated           128.
## 2      2115      1530 2020-03-29 Hypertension Normal           155.
## 3      1545      3840 2020-03-05 Obesity    Stage 1 Hyper~    237.
## 4      6493      1893 2020-09-27 Hypertension Normal           247.
## 5      6388      4426 2020-09-23 Heart Disea~ Stage 2 Hyper~    193.
## 6       412      3595 2020-01-18 Obesity    Elevated           220.
## 7      4202      1062 2020-06-24 Asthma     Elevated           220.
## 8      2361      4672 2020-04-08 Asthma     Elevated           150.
## 9       208      3156 2020-01-09 Hypertension Stage 2 Hyper~    157.
## 10     6670      4573 2020-10-04 Hypertension Elevated           178.
## # i 5,786 more rows
## # i 12 more variables: medication_given <chr>, hospital_location <chr>,
## #   patient_age <dbl>, visit_month <dbl>, patient_age_group <chr>,
## #   cholesterol_risk <chr>, patient_name <chr>, gender <chr>,
## #   admission_date <date>, insurance_status <chr>,
## #   recommended_medication <chr>, dosage_mg <dbl>

# Here is Part 4 of Question 3.
# continued from Part 2 because all diabetes cases are filtered in Part 3 of Question 3
health_record_34 <- health_record_32 %>%
  # count the conditions and call the count number column as patient_count
  count(condition, name = "patient_count") %>%
  # arrange them in a descending order
  arrange(desc(patient_count)) %>%
  # show the top 5 rows
  head(5)
health_record_34

## # A tibble: 5 x 2
##   condition    patient_count
##   <chr>          <int>
## 1 Diabetes        1495
## 2 Obesity         1481
## 3 Hypertension    1461
## 4 Asthma          1436
## 5 Heart Disease   1418
```

Question 4: Exploratory Data Analysis (EDA) (20 points)

Using the cleaned and merged dataset, analyze key healthcare trends:

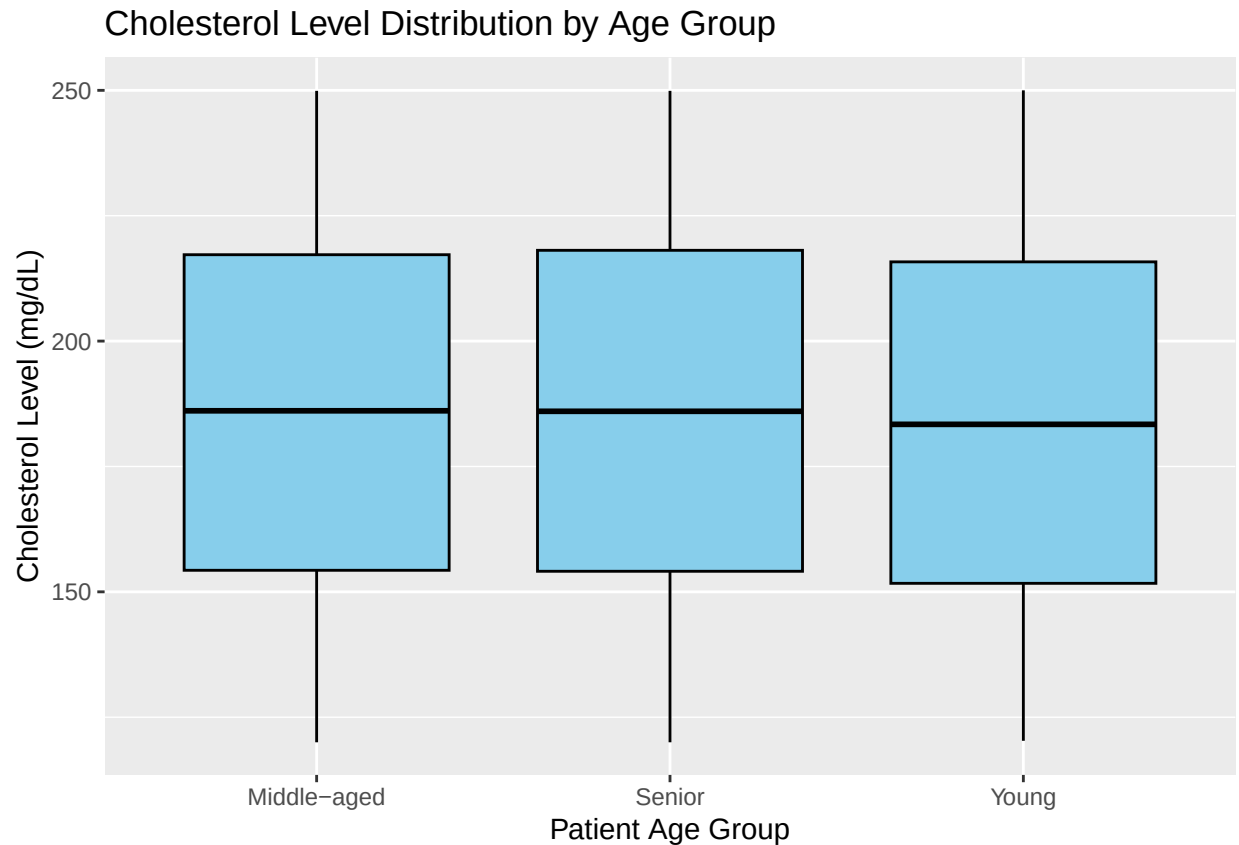
1. What is the average cholesterol level per patient_age_group? Show a summary table and a boxplot.
2. What percentage of patients received medication? Show a bar chart.
3. Which hospital location treated the highest number of patients? Show a summary table.
4. Create a time series plot of total patient visits per month.

Deliverables: - At least 3 visualizations - Summary statistics for key insights - Explanation of findings

```
# Here is Part 1 of Question 4.
# continued from Part 2 of Question 3
cholesterol_41 <- health_record_32 %>%
  # separate patient_age_group with group_by() function
  group_by(patient_age_group) %>%
  # calculate the average cholesterol level with summarise() and mean()
  summarise(average_cholesterol = mean(cholesterol_level, na.rm = TRUE))
cholesterol_41

## # A tibble: 3 x 2
##   patient_age_group average_cholesterol
##   <chr>                <dbl>
## 1 Middle-aged          186.
## 2 Senior               185.
## 3 Young               185.

# draw the boxplot with ggplot()
# the x-axis is patient_age_group and the y-axis is cholesterol_level
ggplot(health_record_33, aes(x = patient_age_group, y = cholesterol_level)) +
  geom_boxplot(fill = "skyblue", color = "black") +
  # name the title, x-axis and y-axis
  labs(title = "Cholesterol Level Distribution by Age Group",
       x = "Patient Age Group",
       y = "Cholesterol Level (mg/dL)")
```

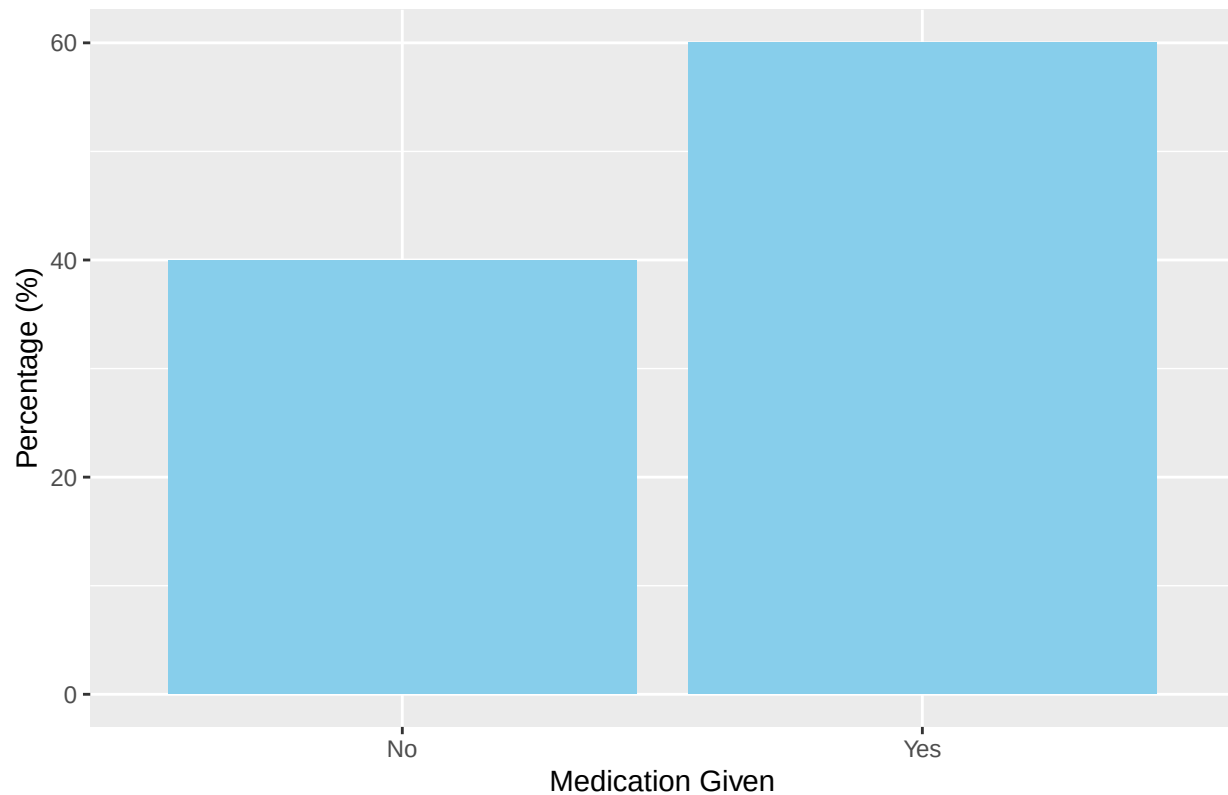


```
# Here is Part 2 of Question 4.
# there is a column called medication_given in health_record_33 tibble
medication_42 <- health_record_32 %>%
  # separate Yes or No in medication_given column with group_by() function
  group_by(medication_given) %>%
  # count patients who received or not
  summarise(patient_count = n()) %>%
  # add a new column to show the percentage of receiving or not
  mutate(percentage = (patient_count / sum(patient_count)) * 100)
medication_42
```

```
## # A tibble: 2 x 3
##   medication_given patient_count percentage
##   <chr>             <int>         <dbl>
## 1 No                2913          40.0
## 2 Yes              4378          60.0
```

```
# draw the bar chart with ggplot()
# the x-axis is medication_given and the y-axis is percentage
ggplot(medication_42, aes(x = medication_given, y = percentage)) +
  # let the Y-axis directly use the values in the data
  geom_bar(stat = "identity", fill = "skyblue") +
  # name the title, x-axis and y-axis
  labs(title = "Percentage of Patients Receiving Medication",
       x = "Medication Given",
       y = "Percentage (%)")
```

Percentage of Patients Receiving Medication



```
# Here is Part 3 of Question 4.
hospital_43 <- health_record_32 %>%
  # separate different cities in hospital_location column with group_by() function
  group_by(hospital_location) %>%
  # count the numbers of treatments in different cities
  summarise(treatment_count = n()) %>%
  # arrange the numbers of treatments in descending order
  arrange(desc(treatment_count))
hospital_43
```

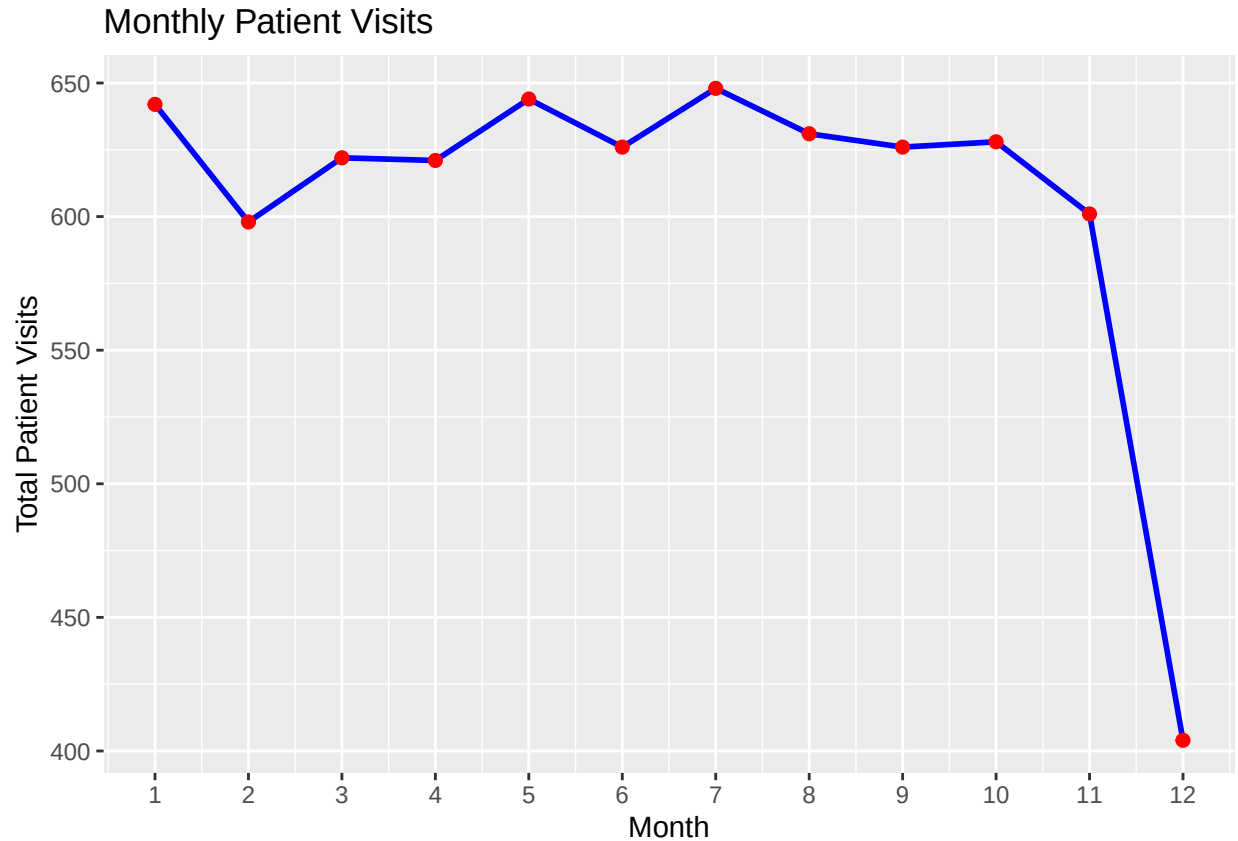
```
## # A tibble: 5 x 2
##   hospital_location treatment_count
##   <chr>                <int>
## 1 Chicago              1542
## 2 LA                   1471
## 3 Houston              1443
## 4 NYC                  1428
## 5 Miami                1407
```

```
# Here is Part 4 of Question 4.
visits_44 <- health_record_32 %>%
  # add a new column to get the month of the visit_date
  # just use month() function in lubridate package
  mutate(visit_month = month(visit_date)) %>%
  # separate different months in visit_month column with group_by() function
  group_by(visit_month) %>%
  # count the total number of visits in each month
```

```
summarise(total_visits = n())
visits_44
```

```
## # A tibble: 12 x 2
##   visit_month total_visits
##       <dbl>       <int>
## 1         1         642
## 2         2         598
## 3         3         622
## 4         4         621
## 5         5         644
## 6         6         626
## 7         7         648
## 8         8         631
## 9         9         626
## 10        10         628
## 11        11         601
## 12        12         404
```

```
# draw the line chart with ggplot()
# the x-axis is visit_month and the y-axis is total_visits
ggplot(visits_44, aes(x = visit_month, y = total_visits)) +
  # show the line
  geom_line(color = "blue", linewidth = 1) +
  # show the points
  geom_point(color = "red", size = 2) +
  # make sure the x-axis scale is just at 1,2,3..., 12
  scale_x_continuous(breaks = 1:12) +
  # name the title, x-axis and y-axis
  labs(title = "Monthly Patient Visits",
        x = "Month",
        y = "Total Patient Visits")
```



Question 5: Healthcare Insights & Recommendations (10 points)

Based on your analysis:

1. Which age group has the highest cholesterol levels? Provide evidence.
2. Which condition is most frequently treated across all hospitals? Explain any patterns found.
3. What recommendations would you give to improve patient health outcomes based on your findings?

Deliverables: - Written answers with supporting evidence - Healthcare recommendations based on findings

1. According to my data analysis in Part 1 of Question 4, middle-aged people tend to have the highest cholesterol levels because they have the highest average cholesterol level (185.7108). However, there is very small difference in cholesterol levels among the three age groups.
2. Generally, diabetes is the most common kind of condition according to Part 4 of Question 3. Specifically, according to Part 4 of Question 2, hypertension is the most frequent one in Chicago, obesity is the most frequent one in Houston, obesity is the most frequent one in LA, asthma is the most frequent one in Miami, and diabetes is the most frequent one in NYC.
3. First of all, all of the diseases analyzed are actually linked to obesity. American society needs to promote healthy eating habits. At the same time, Chicago needs more resources to treat hypertension, Houston and LA need more resources to treat obesity, Miami needs more resources to treat asthma, and NYC needs more resources to treat diabetes.

Question 6: Identifying High-Risk Patients(10 points)

Estimate patient risk levels based on their cholesterol level and blood pressure category.

- Define high_risk as patients with High cholesterol (>200 mg/dL) AND Stage 2 Hypertension.
- Rank the top 10 most at-risk patients based on cholesterol and blood pressure data.
- Provide a scatter plot showing cholesterol level vs. patient age.

Deliverables: - R code for risk level estimation - Ranked list of top 10 high-risk patients - Scatter plot

```
# Here is Parts 1 and 2 of Question 6.
# continued from Part 2 of Question 3
high_risk_patients_61 <- health_record_32 %>%
  # filter rows which satisfy the definition of high-risk
  filter(cholesterol_level > 200 & blood_pressure == "Stage 2 Hypertension") %>%
  # arrange the cholesterol_level in descending order
  # do not need to arrange "Stage 2 Hypertension"
  arrange(desc(cholesterol_level))
# get the top 10 most at-risk patients with head(10)
high_risk_patients_62 <- high_risk_patients_61 %>% head(10)
high_risk_patients_62
```

```
## # A tibble: 10 x 18
##   record_id patient_id visit_date condition blood_pressure cholesterol_level
##   <dbl>      <dbl> <date>      <chr>      <chr>              <dbl>
## 1      3023      3931 2020-05-05 Asthma      Stage 2 Hyper~      250.
## 2      8080      4320 2020-12-02 Asthma      Stage 2 Hyper~      250.
## 3      3947      2981 2020-06-13 Asthma      Stage 2 Hyper~      250.
## 4      1826      1572 2020-03-17 Diabetes    Stage 2 Hyper~      250.
## 5      3588      1945 2020-05-29 Diabetes    Stage 2 Hyper~      250.
## 6      4646      3043 2020-07-12 Asthma      Stage 2 Hyper~      250.
## 7      3621      1172 2020-05-30 Asthma      Stage 2 Hyper~      250.
## 8      8338      1785 2020-12-13 Hypertension Stage 2 Hyper~      250.
## 9      7051      1864 2020-10-20 Hypertension Stage 2 Hyper~      250.
## 10     1990      3279 2020-03-23 Heart Disea~ Stage 2 Hyper~      249.
## # i 12 more variables: medication_given <chr>, hospital_location <chr>,
## #   patient_age <dbl>, visit_month <dbl>, patient_age_group <chr>,
## #   cholesterol_risk <chr>, patient_name <chr>, gender <chr>,
## #   admission_date <date>, insurance_status <chr>,
## #   recommended_medication <chr>, dosage_mg <dbl>
```

```
# Here is Part 3 of Question 6.
# continued from Part 2 of Question 6
# draw the scatter plot with ggplot()
ggplot(high_risk_patients_62, aes(x = patient_age, y = cholesterol_level)) +
  # the x-axis is patient_age and the y-axis is cholesterol_level
  geom_point(data = high_risk_patients_62, aes(x = patient_age, y = cholesterol_level), color = "red", size = 100)
# name the title, x-axis and y-axis
labs(title = "Cholesterol Level vs. Patient Age",
      x = "Patient Age",
      y = "Cholesterol Level (mg/dL)")
```




Final Submission Requirements PDF file containing all responses, explanations, tables, and visualizations. .R script file with well-commented, reproducible code. Submit your pdf onto Gradescope. We will request your R script file if necessary.

Final Notes You should explain your method or justification in a paragraph(s) for each sub questions. Proofread before submitting. Ensure all R code runs without errors. Ask questions early if anything is unclear.

R Appendix

```
knitr::opts_chunk$set(echo = TRUE,eval = TRUE) # when you knit for submitting, change eval = FALSE to TRUE
library(tidyverse)
# import the datasets

# messy_health_records.csv -> messy_health_records
messy_health_records <- read_csv("messy_health_records.csv")

# patients.csv -> patients
patients <- read_csv("patients.csv")

# medications.csv -> medications
medications <- read_csv("medications.csv")
# briefly know the structures of the three csv files with head()
head(messy_health_records)
```

```

head(patients)
head(medications)
# Here is Part 1 of Question 1.
# check for missing values by counting the numbers of missing values in all columns
missing_values_summary_11 <- colSums(is.na(messy_health_records))
missing_values_summary_11

# Here is Part 2 of Question 1.
# use drop_na() to remove those rows with missing values
# there are thousands of records, it is okay to drop those rows with NA
messy_health_records_12 <- messy_health_records %>% drop_na()
messy_health_records_12

# Here is Part 3 of Question 1.
# use ymd() in lubridate package to convert visit_date into Date format
messy_health_records_13 <- messy_health_records_12 %>%
  mutate(visit_date = ymd(visit_date))
messy_health_records_13

# Here is Part 4 of Question 1.
# remove those repetitive rows by using distinct()
messy_health_records_14 <- messy_health_records_13 %>% distinct()
messy_health_records_14
# count the number of total rows after finishing Part 3 with nrow()
row_count <- nrow(messy_health_records_13)
# count the number of unique rows after finishing Part 3 with n_distinct()
unique_row_count <- n_distinct(messy_health_records_13)
# the number of duplicate rows is just the difference between row_count and unique_row_count
duplicates <- row_count - unique_row_count
duplicates

# Here is Part 5 of Question 1.
# calculate the upper and lower quantiles of cholesterol_level with quantile()
# the lower quartile is 0.25
cho25 <- quantile(messy_health_records_14$cholesterol_level, 0.25, na.rm = TRUE)
# the upper quartile is 0.75
cho75 <- quantile(messy_health_records_14$cholesterol_level, 0.75, na.rm = TRUE)
# Interquartile range = upper quartile - lower quartile
IQR_cho <- cho75 - cho25
# consider the outliers as 1.5 interquartile range smaller than cho25 or greater than cho75
lower_bound_cho <- cho25 - 1.5 * IQR_cho
upper_bound_cho <- cho75 + 1.5 * IQR_cho

# calculate the upper and lower quantiles of patient_age with quantile()
# the lower quartile is 0.25
age25 <- quantile(messy_health_records_14$patient_age, 0.25, na.rm = TRUE)
# the upper quartile is 0.75
age75 <- quantile(messy_health_records_14$patient_age, 0.75, na.rm = TRUE)
# Interquartile range = upper quartile - lower quartile
IQR_age <- age75 - age25
# consider the outliers as 1.5 interquartile range smaller than cho25 or greater than cho75
lower_bound_age <- age25 - 1.5 * IQR_age
upper_bound_age <- age75 + 1.5 * IQR_age

```

```

# use filter() to remain those ordinary records
messy_health_records_15 <- messy_health_records_14 %>%
  # normal cholesterol_level should be between lower_bound_cho and upper_bound_cho
  filter(cholesterol_level >= lower_bound_cho & cholesterol_level <= upper_bound_cho) %>%
  # normal patient_age should be between lower_bound_age and upper_bound_age
  filter(patient_age >= lower_bound_age & patient_age <= upper_bound_age)
messy_health_records_15
# Here is Part 1 of Question 2.
# continued from Part 5 of Question 1
messy_health_records_21 <- messy_health_records_15 %>%
  # use month() in lubridate package and mutate() to add the new column
  mutate(visit_month = month(visit_date))
messy_health_records_21

# Here is Part 2 of Question 2.
# add the new column called "patient_age_group" with mutate()
messy_health_records_22 <- messy_health_records_21 %>%
  # there are 3 cases. Use case_when() to separate the 3 cases.
  mutate(patient_age_group = case_when(
    patient_age >= 18 & patient_age <= 35 ~ "Young",
    patient_age >= 36 & patient_age <= 60 ~ "Middle-aged",
    patient_age >= 61 ~ "Senior"))
messy_health_records_22

# Here is Part 3 of Question 2.
# similarly, add the new column called "cholesterol_risk" with mutate()
messy_health_records_23 <- messy_health_records_22 %>%
  # similarly, there are 3 cases. Use case_when() to separate the 3 cases.
  mutate(cholesterol_risk = case_when(
    cholesterol_level < 160 ~ "Low",
    cholesterol_level >= 160 & cholesterol_level <= 200 ~ "Moderate",
    cholesterol_level > 200 ~ "High"))
messy_health_records_23

# Here is Part 4 of Question 2.
messy_health_records_24 <- messy_health_records_23 %>%
  # per hospital location and condition means we need to use group_by() to separate
  group_by(hospital_location, condition) %>%
  # count the number of patients per hospital with the same condition
  summarize(visit_count = n()) %>%
  # change the NA to 0 with replace_na()
  # mutate should be used to help keep the list structure
  mutate(visit_count = replace_na(visit_count, 0)) %>%
  # convert the data frame to wider form with pivot_wider()
  # names_from and values_from are two parameters
  pivot_wider(names_from = condition, values_from = visit_count)
messy_health_records_24
# Here is Part 1 of Question 3.
# continued from Part 3 of Question 2
health_record_31 <- messy_health_records_23 %>%
  # use left_join() to join the two datasets according to the "patient_id" column
  left_join(patients, by = "patient_id")
health_record_31

```

```

# Here is Part 2 of Question 3.
health_record_32 <- health_record_31 %>%
  # use left_join() to join the two datasets according to the "condition" column
  left_join(medications, by = "condition")
health_record_32

# Here is Part 3 of Question 3.
health_record_33 <- health_record_32 %>%
  # use filter() to remain those rows we need here
  filter(dosage_mg <= 200)
health_record_33

# Here is Part 4 of Question 3.
# continued from Part 2 because all diabetes cases are filtered in Part 3 of Question 3
health_record_34 <- health_record_32 %>%
  # count the conditions and call the count number column as patient_count
  count(condition, name = "patient_count") %>%
  # arrange them in a descending order
  arrange(desc(patient_count)) %>%
  # show the top 5 rows
  head(5)
health_record_34

# Here is Part 1 of Question 4.
# continued from Part 2 of Question 3
cholesterol_41 <- health_record_32 %>%
  # separate patient_age_group with group_by() function
  group_by(patient_age_group) %>%
  # calculate the average cholesterol level with summarise() and mean()
  summarise(average_cholesterol = mean(cholesterol_level, na.rm = TRUE))
cholesterol_41

# draw the boxplot with ggplot()
# the x-axis is patient_age_group and the y-axis is cholesterol_level
ggplot(health_record_33, aes(x = patient_age_group, y = cholesterol_level)) +
  geom_boxplot(fill = "skyblue", color = "black") +
  # name the title, x-axis and y-axis
  labs(title = "Cholesterol Level Distribution by Age Group",
       x = "Patient Age Group",
       y = "Cholesterol Level (mg/dL)")

# Here is Part 2 of Question 4.
# there is a column called medication_given in health_record_33 tibble
medication_42 <- health_record_32 %>%
  # separate Yes or No in medication_given column with group_by() function
  group_by(medication_given) %>%
  # count patients who received or not
  summarise(patient_count = n()) %>%
  # add a new column to show the percentage of receiving or not
  mutate(percentage = (patient_count / sum(patient_count)) * 100)
medication_42

# draw the bar chart with ggplot()
# the x-axis is medication_given and the y-axis is percentage
ggplot(medication_42, aes(x = medication_given, y = percentage)) +
  # let the Y-axis directly use the values in the data

```

```

geom_bar(stat = "identity", fill = "skyblue") +
# name the title, x-axis and y-axis
labs(title = "Percentage of Patients Receiving Medication",
      x = "Medication Given",
      y = "Percentage (%)")

# Here is Part 3 of Question 4.
hospital_43 <- health_record_32 %>%
  # separate different cities in hospital_location column with group_by() function
  group_by(hospital_location) %>%
  # count the numbers of treatments in different cities
  summarise(treatment_count = n()) %>%
  # arrange the numbers of treatments in descending order
  arrange(desc(treatment_count))
hospital_43

# Here is Part 4 of Question 4.
visits_44 <- health_record_32 %>%
  # add a new column to get the month of the visit_date
  # just use month() function in lubridate package
  mutate(visit_month = month(visit_date)) %>%
  # separate different months in visit_month column with group_by() function
  group_by(visit_month) %>%
  # count the total number of visits in each month
  summarise(total_visits = n())
visits_44

# draw the line chart with ggplot()
# the x-axis is visit_month and the y-axis is total_visits
ggplot(visits_44, aes(x = visit_month, y = total_visits)) +
  # show the line
  geom_line(color = "blue", linewidth = 1) +
  # show the points
  geom_point(color = "red", size = 2) +
  # make sure the x-axis scale is just at 1,2,3..., 12
  scale_x_continuous(breaks = 1:12) +
  # name the title, x-axis and y-axis
  labs(title = "Monthly Patient Visits",
        x = "Month",
        y = "Total Patient Visits")

# Here is Parts 1 and 2 of Question 6.
# continued from Part 2 of Question 3
high_risk_patients_61 <- health_record_32 %>%
  # filter rows which satisfy the definition of high-risk
  filter(cholesterol_level > 200 & blood_pressure == "Stage 2 Hypertension") %>%
  # arrange the cholesterol_level in descending order
  # do not need to arrange "Stage 2 Hypertension"
  arrange(desc(cholesterol_level))
# get the top 10 most at-risk patients with head(10)
high_risk_patients_62 <- high_risk_patients_61 %>% head(10)
high_risk_patients_62

# Here is Part 3 of Question 6.
# continued from Part 2 of Question 6

```

```

# draw the scatter plot with ggplot()
ggplot(high_risk_patients_62, aes(x = patient_age, y = cholesterol_level)) +
  # the x-axis is patient_age and the y-axis is cholesterol_level
  geom_point(data = high_risk_patients_62, aes(x = patient_age, y = cholesterol_level), color = "red", )
  # name the title, x-axis and y-axis
  labs(title = "Cholesterol Level vs. Patient Age",
        x = "Patient Age",
        y = "Cholesterol Level (mg/dL)")

```